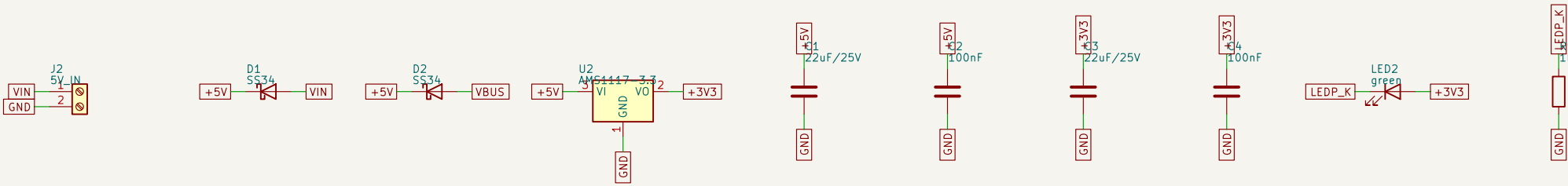
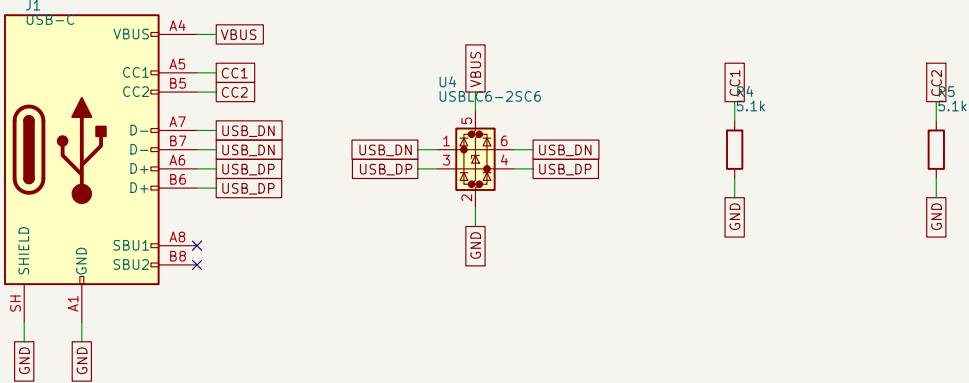


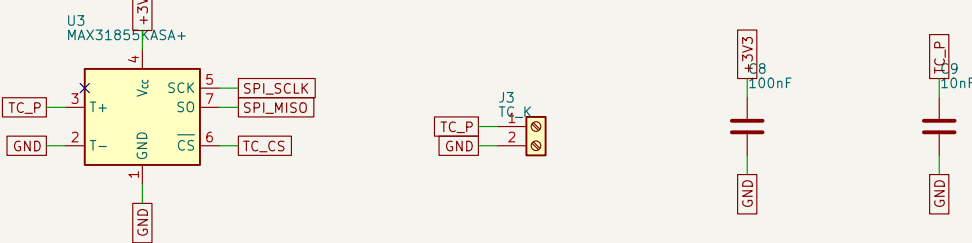
POWER IN (5V DC terminal or USB, ORed Schottky diodes)



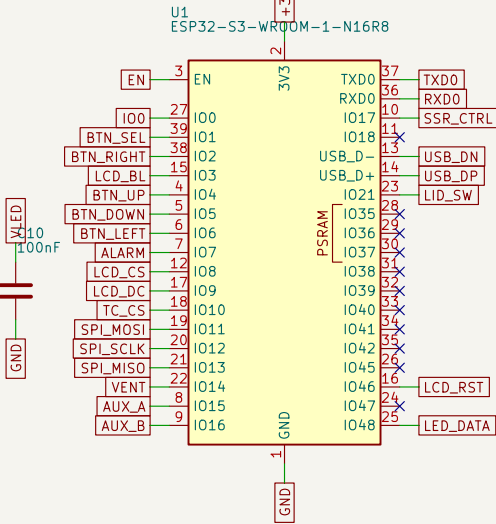
USB-C (native USB flashing + ESD)



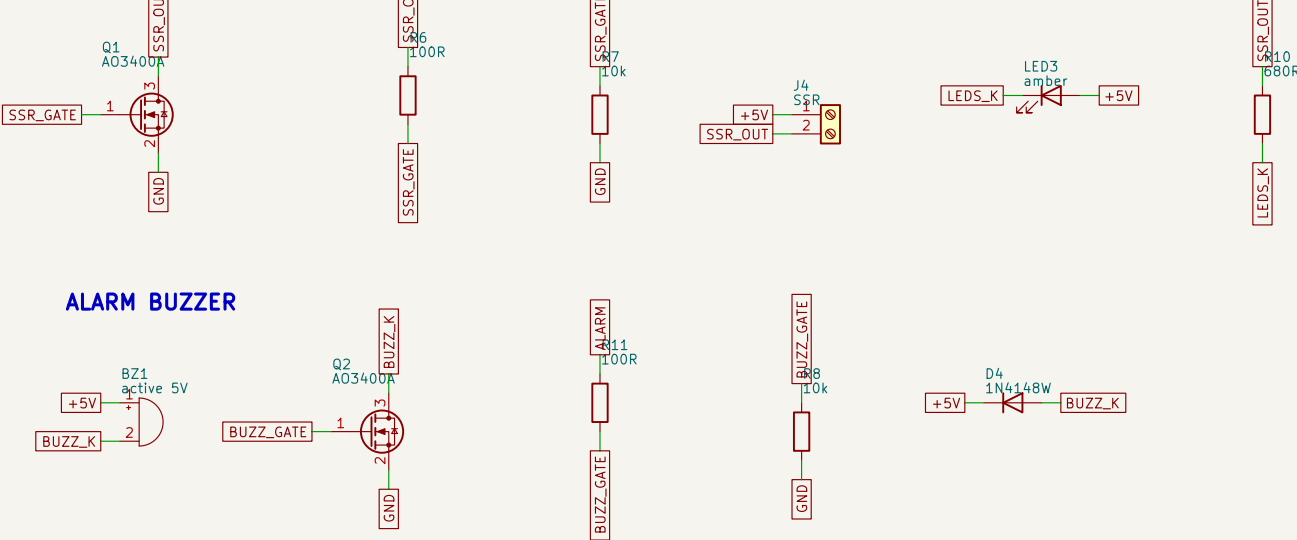
THERMOCOUPLE MAX31855 (T- grounded per datasheet)



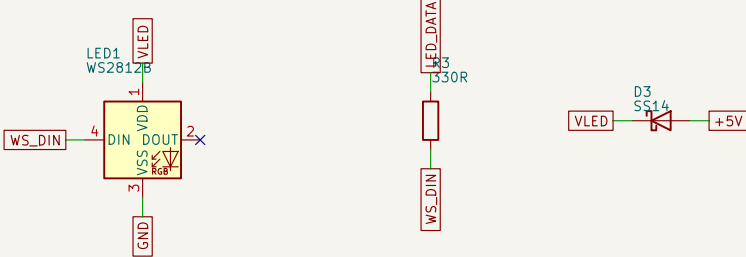
ESP32-S3-WROOM-1 (GPIOs = firmware Kconfig defaults)



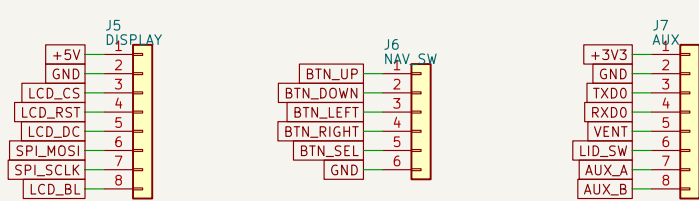
SSR DRIVE (low-side switch J4 supplies the +5V loop)



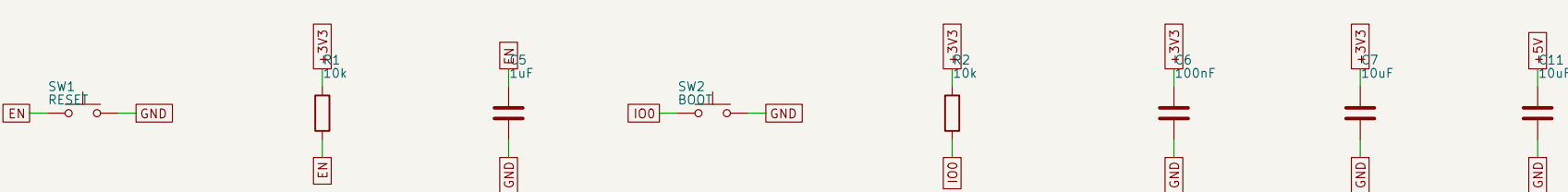
WS2812B STATUS LED (VDD dropped ~4.6V for 3.3V data margin)



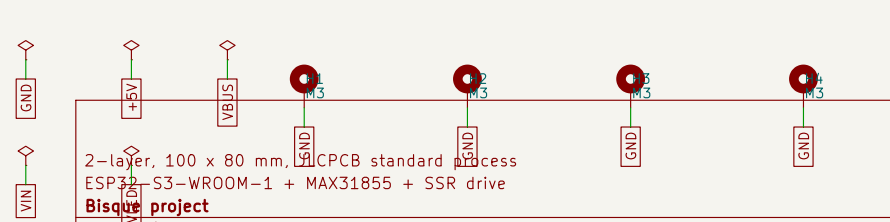
HEADERS DISPLAY(J5) NAV(J6) AUX(J7)



RESET / BOOT / DECOUPLING



MOUNTING / POWER FLAGS



Bisque kiln controller - ESP32-S3-WROOM-1
SSR terminal J4: +5V / SSR- loop, switched low-side by Q1 (boot-safe: R7 pulldown)
TC terminal J3: pin1 = K+ (yellow), pin2 = K- (red, grounded at U3)
Nav switch J6 is panel-mounted; inputs use ESP32 internal pull-ups
Display 15 pinout: 3V3 GND ICS RST DC MOSI SCK BL (ST7796S SPI module)

2-layer, 100 x 80 mm, L1PCB standard process		
ESP32-S3-WROOM-1 + MAX31855 + SSR drive		
Bisque project		
Sheet: /		
File: bisque-controller.kicad_sch		
Title: Bisque Kiln Controller		
Size: A3	Date: 2026-07-20	Rev: A
KiCad E.D.A. 10.0.5		Id: 1/1